BUS 414/BUS 410

Linear Programming 1: Introduction

Introduction

Linear Programming has proved to be one of the most widely used quantitative decision making tools. It was developed as a major technique in solving military logistical problems and has since been deployed for commercial purposes in a wide range of sectors. Today, linear programming is increasingly being used to address decision making problems in agriculture, production planning and scheduling, investment planning, corporate planning, transportation arrangement, assignment problems, distribution management, advertising, and numerous other areas.

Learning Outcomes

When you have studied this session, you should be able to:

- List and explain the major linear programming terms and convention
- List and explain the assumptions governing the use of linear programming.
- List the types of decision problems linear programming can be used to solve.
- List and explain the various terminologies (terms) in linear programming.
- Give the general form of a linear programming problem.
- State the matrix form of a linear programming problem.
- State the advantages of using linear programming.
- State the limitations of linear programming.
- State the steps involved in converting worded allocation of resource.
- Briefly describe the systematic formulation of linear programming problem.
- Transform simple real-life problems to linear programming problems.
- Solve a linear programming problem with two decision variables by graphical method.
- Transform a linear programming problem to its standard form and its canonical format.
- Recall the terms that are used in the Simplex solution procedure.
- Draw and fill in the inputs or parameters of an LP into the initial Simplex tableau.
- Know how to select the pivot row and pivot column.
- Perform Simplex iterations.

Definition

Linear programming is a mathematical methodology of allocating scarce (limited) resources to uses in such a way that will best achieve the objective of the decision maker. It employs mathematical expressions to model or program a problem situation. It is a constrained optimization technique which seeks the achievement of a single objective such as maximizing or minimizing a criterion in a model. The criterion could be profit or revenue related in a maximization problem or cost or time related in a minimization problem. However, in the real world, resources are limited. Therefore attaining the objective of maximizing profit or minimizing cost is constrained by the limitation of available resources.

NOTE: Linear programming technique can only be applied within an all-linear framework (that is, a situation where the unknown variables in the criterion function and the constraints inequalities or equations are all linear) under a condition of certainty.

Linear Programming Terms and Convention

The following are the definitions of the several commonly used linear programming terms and conventions:

Decision Variables: A set of unknown quantities, represented in the linear programming model by mathematical symbols, whose values are to be determined.

Side (structural or System) Constraints: A set of inequalities/equations that express the relevant limitations or restrictions, in terms of availability of resources, in the problem situation being modelled mathematically. These limitations or restrictions determine the extent to which a resource can be used.

Right-Hand-Side Values: These are values that express resource levels that we seek to utilize. The RHS value is usually a constant represented mathematically by the symbol b_i, where i denotes the ith resource. The term RHS value originates from the positioning of the value on the right-hand-side of each constraint inequality or equation.

Slack Variables: These are variables that express the possibility of positive deviation of the sum of the terms on the left-hand-side of a constraint inequality from the right-hand side value. Slack variables are represented in the linear programming by the symbol S_i, where i denotes the ith resource (i = 1, 2, 3,…, I) and I is the number of side constraints with positive deviations.

Surplus Variables: These are variables that express the possibility of negative deviation of the sum of the terms on the left-hand-side of a constraint inequality from the right-hand side value. Surplus variables are represented in the linear programming by the symbol S_i, where i denotes the ith resource (i = 1, 2, 3, …, I) and I is the number of side constraints with negative deviations.

Technological Coefficients: These are numerical values or constants, represented by a_{ij}, that express the per unit usage of the RHS resource (b_i) by each decision variable X_j.

Contribution Coefficients: These are numerical values or constants, represented by C_j, that express the per unit contribution of each decision variable X_j to the total (or overall) profit.

Cost Coefficients: These are numerical values or constants, represented by C_j, that express the per unit cost of each decision variable X_j to the total (or overall) cost.

Z (Value): An unknown value that we seek to optimize either by maximization (that is, making it as large as possible as in profit oriented situation) or minimization (that is, making it as small as possible as in cost oriented situations).

Objective (or Criterion) Function: This is a mathematical expression that equates **Z** to the sum of the product of individual contribution (or cost) coefficient and the respective decision variable in a problem situation.

Tableau: It is a specialized matrix of numbers used in the simplex solution procedure.

Basic Variables: A set of variables, which could be either decision or slack variables, that exist in the basis column of the simplex tableau.

Non-basic Variables: These are all those variables, either decision or slack variables, that are not in the basis column of the simplex tableau. These variables are also called zero-variables as their value at the optimal solution is zero.

Linear Programming Assumptions

Linear programming model are governed by the following assumptions:

- **Linearity**: The decision variables must appear only in linear relationships in both the objective function and constraint inequalities/equations. In general, a linear relationship involving k decision variables X_1, X_2, X_3, X_4, …, X_k is an expression of the form $C_1X_1 + C_2X_2 + C_3X_3 + C_4X_4 + \ldots + C_kX_k$, where C_1, C_2, C_3, C_4, …, C_k are k given numbers or constants which could be either positive or negative.

- **Additive:** The additive condition requires that the terms in the objective or criterion function (total measure of outcome or performance) and the constraints (the total sum of resource usage) must be additive. The assumption that the terms of each relationship are additive implies that the activities are independent. By this is meant that cross-product terms of decision variables, say X_1X_2, as a result of interaction between some of the variables, do not arise.

- **Divisibility:** The divisibility assumption requires that the solution values obtained for decision variables, X_j, must not be restricted to integer values. This implies that the decision variables can take on any fractional solution value. Hence, they are referred

to as continuous.

- **Deterministic:** This assumption requires that the model parameters (C_j, a_{ij}, and b_i) in linear programming should be known constants. This assumes a decision problem in a static time frame in which all parameters are known with certainty. In real life situations however, model parameters are rarely deterministic since they reflect the future as well as the present conditions, and it is somehow difficult to know the future conditions with certainty.

There, however, exist several ways to cope with parameter uncertainty in linear programming models. One of these methods is by using the chance-constrained programming techniques, a method that incorporates probabilistic occurrences in model parameters for the future based on scenario development. In the alternative, however, sensitivity analysis may be used. Sensitivity analysis is a technique developed to test solution values to ascertain how sensitive they are to parameter changes.

- **Non-Negativity**: This assumption restricts decision variables from having negative values since activities in real world situation cannot be negative. By this is meant that scarce resources cannot be committed to producing negative or imaginary products. Resources are either used in producing real products or they are left idle.

Areas of Application

The areas of application of linear programming include, but not limited to, the following:

- **Investment Planning**: Financial mix, project selection, and portfolio selection
- **Transportation Arrangement**: Cargo shipment/loading, facilities expansion, how to ship goods to customers at least cost, school enrolment and political re-districting.
- **Distribution Management**: Selecting the most economical distribution network
- **Production Planning (Product Mix)**: Deciding what goods are to be produced and in what quantity
- **Production Scheduling**: Deciding which jobs should be done on which machines
- **Assignment Problem**: Matching people to jobs, work to machines and contracts to bidders
- **Adverting/Promotional Planning (Media Mix)**: Selecting which advertising media will create the largest exposure to targeted audience
- **Corporate Planning**: Mapping out the best strategy for entire company operations

- **State Security**: Strategic military effectiveness analysis
- **Blending**: Product blend and dietary problems.
- **Military Applications**: Selecting an air weapon system against enemy so as to keep them pinned down and at the same time minimize the amount of aviation gasoline used.
- **Agricultural Applications**: In the form of farm economics and farm management. Farm economics deals with agricultural economy of a region or nation, while farm management is concerned with the problems of the individual farm.

The General Formulation of Linear Programming Problem

Suppose we need to find the values of n decision variables $X_1, X_2, X_3, \ldots, X_n$ that will maximize (or minimize) a given objective function denoted by Z as given below. The general linear programming problem is stated as:

$$\text{Maximize (or minimize) } \mathbf{Z} = C_1X_1 + C_2X_2 + C_3X_3 + \ldots + C_nX_n \qquad (1.1)$$

Subject to the m-constraints

$$\begin{aligned}
&a_{11}X_1 + a_{12}X_2 + a_{13}X_3 + \ldots + a_{1n}X_n \; [\leq = \geq] \; b_1 \\
&a_{21}X_1 + a_{22}X_2 + a_{23}X_3 + \ldots + a_{2n}X_n \; [\leq = \geq] \; b_2 \\
&a_{31}X_1 + a_{32}X_2 + a_{33}X_3 + \ldots + a_{3n}X_n \; [\leq = \geq] \; b_3 \\
&\quad . \qquad . \qquad . \qquad . \qquad . \\
&a_{i1}X_1 + a_{i2}X_2 + a_{i3}X_3 + \ldots + a_{in}X_n \; [\leq = \geq] \; b_i \qquad (1.2) \\
&\quad . \qquad . \qquad . \qquad . \qquad . \\
&a_{m1}X_1 + a_{m2}X_2 + a_{m3}X_3 + \ldots + a_{mn}X_n \; [\leq = \geq] \; b_m
\end{aligned}$$

Note that the constraints may take the inequality form $\leq$ or $\geq$ or even equation form.

Finally, the n variables must satisfy the non-negativity requirement

$$X_j \geq 0 \text{ for } j = 1, 2, \ldots, n \qquad (1.3)$$

Matrix Form of Linear Programming Problem

The above problem is stated in its matrix equivalent as

$$\text{Maximize (or Minimize) } \mathbf{Z} = CX \qquad \text{Criterion function}$$

such that

$$AX \; [\leq = \geq] \; b \qquad \text{Constraint inequalities or equations}$$

with $b > 0$, $X \geq 0$ Non-negativity restriction

where $X = [X_1, X_2, X_3, \ldots, X_n]^T$; $C = (C_1, C_2, C_3, \ldots, C_n)$; $b = (b_1, b_2, b_3, \ldots, b_m)$

$$A = \begin{matrix} a_{11} & a_{12} & a_{13} & \ldots. & a_{1n} \\ a_{21} & a_{22} & a_{23} & \ldots. & a_{2n} \\ a_{31} & a_{32} & a_{33} & \ldots. & a_{3n} \\ . & . & . & & . \\ a_{m1} & a_{m2} & a_{m3} & \ldots. & a_{mn} \end{matrix}$$

Advantages of using Linear Programming

The advantages of using linear programming include:

i) It helps in attaining the optimum use of productive resources, as well as indicating how a decision-maker can employ his productive factors effectively by selecting and distributing these resources.

ii) The technique improves the quality of decisions by making the decision-making approach of the user to become more objective and less subjective.

iii) Linear programming technique provides possible and practical solutions to real world problems. However, since there might be other constraints operating outside a problem which must be taken into account, necessary modification of the technique is required in order to allow some flexibility for the decision maker. For instance, the fact that a firm can produce so many units of a product does not mean that all can be sold. By modifying the constraints such issues can be accommodated in the linear programming problem.

iv) The most significant advantage of linear programming is in the highlighting of bottlenecks in the production process. This is important because when a bottleneck occurs, some machines may not be able to meet demand while other machines remain idle.

v) Linear programming helps in re-evaluation (sensitivity analysis) of a basic plan for changing conditions. If conditions change when the plan is partly carried out, they can be determined so as to adjust the remainder of the plan for best results.

Limitations of Linear Programming

Notwithstanding the many advantages and wide areas of application, there are some limitations associated with the technique, some of which are:

- The technique treats all relationships among decision variables as linear. In real -life situations concerning business and industrial problems, neither the

objective function nor the constraints are linearly related to the variables.

- Linear programming technique does not guarantee integer valued solutions. For instance, in finding the optimal number of police personnel to be deployed to a troubled area to curb a disturbance, a non-integer solution will be meaningless. Rounding off the solution to the nearest integer may not yield an optimal solution, hence in such a case, integer programming is used to ensure integer value to the decision variables.

- Linear programming model does not take into consideration the effect of time and uncertainty. Thus, the linear programming model should be defined in such a way that any change due to internal as well as external factors can be incorporated.

- The parameter inputs in the model are assumed to be constant, whereas in real-life situations they are frequently neither constant nor known.

- Linear programming deals with only single objective, whereas in real-life situations we may have more than one objective which sometimes may be conflicting.

Steps in formulating a linear programming problem

The steps involved in formulating a real-life problem as linear programming are as follows:

Step 1: Write an accurate description of the problem and identify the various variables of the real system.

Step 2: Identify the parameters such as technological coefficients, contribution or cost coefficients and resource availability (right-hand-side values of each resource)

Step 3: Define the decision variables clearly, using mathematical symbol to denote or represent each of the variables.

Step 4: Express the mathematical statement of the decision-maker in terms of the decision variables and contribution or cost coefficient.

Step 5: Express each constraint limiting the achievement of the decision-maker's objective mathematically in terms of the decision variable, technological coefficient, and limitation of the availability of each resource.

Systematic formulation of linear programming problem.

In general, many things can go wrong in the formulation of a real-life problem as a linear programming. To be on the safe side, one should make it a habit to ask the following questions:

Is enough information provided in step 1?
Is it possible to determine or estimate all the costs, profits and other parameters for the LP as required by step 2?
In steps 3, 4 and 5, is it often possible that the choice of the decision variables (step 3) will be the best choice for expressing the objective function and/or constraints (steps 4 and 5)? That is to say, is there no other choice of decision variables that might lead to an easier LP formulation? The point being made here is that steps 3, 4 and 5 often interact with each other to arrive at the final formulation; hence the initial choice of decision variables in step 3, oftentimes, may not work at steps 4 and 5. One must take note of the divisibility assumption in step 3; while ensuring linearity assumption is implicit in steps 4 and 5.

Transformation of simple real-life problems to linear programming problem.

Problem 1

B & G Inc. manufactures two products that require two components, A and B. One unit of product 1 requires 3 pieces of component A, 2 pieces of component B, and 1 hour of labour. Each unit of product 2 requires 2 pieces of component A, 4 pieces of component B and 1 hour of labour. Only 18 pieces and 20 pieces of components A and B, respectively, are available for the production of the two products. However, the production manager must utilize at least 2 hours in the production of the two products. Demand forecast also shows that not more than 4 units of product 2 should be produced. The profit contribution of each unit of product 1 is ₦3.00, while a unit of product 2 contributes ₦4.00 to profit. Formulate the problem as a linear program.

Problem Formulation

Decision variables: Let,

x_1 = number of units of product 1 to be produced.
x_2 = number of units of product 2 to be produced

The mathematical statement of the problem is:

Maximize $Z = 3x_1 + 4x_2$ (Profit, ₦)

Subject to:

$3x_1 + 2x_2 \leq 18$ (Availability of component A)
$2x_1 + 4x_2 \leq 20$ (Availability of component B)
$x_1 + x_2 \geq 2$ (Labour hours)
$x_2 \leq 4$ (Limitation on product 2)
$x_1, x_2 \geq 0$ (Non-negativity)

Problem 2

A company produces two products, A and B. Each product has resource requirements and profit contribution as follows:

Resource	Product A	Product B	Total Resource Availability
Material (Kg/unit)	1	2	10Kg
Labour (hr/unit)	6	6	36 hours
Profit (N/unit)	4	5	

In addition, demand forecasts have shown that at least 5 units of product B should be produced to meet the demand of customers. What product mix will make the company's profit as large as possible?

Formulation

The mathematical statement of the LP problem is as follows:

Decision Variables

Let X_j denote the number of product j to produce (j = A, B)

Objective Function

Since the problem relates to profit making, the objective of the company will be to maximize the total profit to be derived from the production of the two products. Thus we have:

Maximize: $Z = 4X_A + 5X_B$

Subject to:

$X_A + 2X_B \leq 10$ [Material availability]

$6X_A + 6X_B \leq 36$ [Labour hours]

$X_B \geq 5$ [production of product B]

$X_A , X_B \geq 0$

Problem 3

A farmer, who raises livestock, is mixing a feed from two different ingredients, A and B. Each bag of ingredient A contributes a pound of the required vitamins and four pounds of the required minerals to the feed mix. Each bag of ingredient B contributes two pounds of the required vitamins and two pounds of the required minerals to the feed mix. The feed mix must contain at least 40 pounds of vitamins and 60 pounds of minerals. Because of the additional nutrients in the ingredients, each batch of feed mix must contain at least 5 bags of each ingredient. A bag of ingredient A costs N15, while a bag of ingredient B costs N20. The farmer would like to know the number of bags of each ingredient to use in the feed mix so as to minimize total cost.

Formulation

Decision Variables

Let X_j denote the number of bags of ingredient j (j = A, B) to use for the feed mix

Objective Function

The objective function is stated as:

Minimize: $Z = 15X_A + 20X_B$

Subject to:

$X_A + 2X_B \geq 40$

$4X_A + 2X_B \geq 60$

$X_A \geq 5$

$X_B \geq 5$

$X_j \geq 0$ for j = A, B.

Problem 4

A production manager wants to determine the production schedule of a product for the next five months. Past production records showed that only 2000 units of the product can be produced per month on regular time basis. An additional 600 units can be produced monthly on an overtime basis. The unit cost of items produced on regular time basis is N75 while an item produced on overtime basis costs N86. Contracted sales per month for the next five months are as follows:

Month	Contracted sales (Units)
1	1200
2	2100
3	2400
4	3000
5	4000

Inventory carrying cost is N15 per unit per month. The manager does not want any inventory carried over past the fifth month. The manager would like to determine the monthly production level that will minimize the total production and inventory cost. Formulate the problem as a linear program.

Formulation

Decision Variables: Let X_{it} denote the number of units of the product produced by using mode i (i = r [regular], o [overtime]) in month t (t = 1, 2, 3, 4, 5); and I_t (t = 1, 2, 3, 4) denote the number of units of the product carried as inventory at the end of month t into month (t+1).

Objective Function

Minimize: $Z = N\{75(X_{r1} + X_{r2} + X_{r3} + X_{r4} + X_{r5}) + 86(X_{o1} + X_{o2} + X_{o3} + X_{o4} + X_{o5}) + 15(I_1 + I_2 + I_3 + I_4)\}$

Subject to:

$X_{rt} \leq 2000$ for t = 1, 2, 3, 4, 5	regular time production constraint
$X_{ot} \leq 600$ for t = 1, 2, 3, 4, 5	overtime production constraint

Contracted sales constraints

$X_{r1} + X_{o1} - I_1 = 1200$	sales constraint for month 1
$X_{r2} + X_{o2} + I_1 - I_2 = 2100$	sales constraint for month 2
$X_{r3} + X_{o3} + I_2 - I_3 = 2400$	sales constraint for month 3
$X_{r4} + X_{o4} + I_3 - I_4 = 3000$	sales constraint for month 4
$X_{r5} + X_{o5} + I_4 = 4000$	sales constraint for month 5

$X_{it} \geq 0$ for I = r, o and t = 1, 2, 3, 4, 5

$I_t \geq 0$ for t = 1, 2, 3, 4.

Problem 5

A manufacturing company is engaged in producing three types of products 1, 2 and 3. The production department produces, each day, components sufficient to make 45 units of product 1, 20 units of product 2 and 35 units of product 3. The management is confronted with the problem of optimizing the daily production of products in assembly department where only 125 man-hours are available daily to assemble the three products. The assembly time required and the profit contribution per unit of each of the products is presented in the table below.

Product	Assembly Time per Product(hours)	Profit Contribution per unit of Product (N)
1	0.75	15
2	1.5	22
3	2.3	40

The company has a daily order commitment for 25 units of product 1 and a total unit of 20 units of products 2 and 3. Formulate the problem as a linear program to maximize the total profit.

Problem Formulation

Decision Variables: Let X_j be the number of units of product j (j = 1, 2, 3) to be produced.

Objective Function

Maximize: $Z = 15X_1 + 22X_2 + 40X_3$

Subject to the constraints

(i) Labour and materials constraints

$$0.75X_1 + 1.5X_2 + 2.3X_3 \leq 125$$

$$X_1 \leq 45$$

$$X_2 \leq 20$$

$$X_3 \leq 35$$

(ii) Order commitment constraint

$$X_1 \geq 25$$

$$X_2 + X_3 \geq 20$$

$$X_1, X_2, X_3 \geq 0$$

Graphical Solution of Linear Programming Problem

This session discusses the two methods of solving a linear programming problem, namely: the graphical method and the Simplex method. The graphical method is used when a linear programming problem has just two decision variables; while the Simplex method is used to address a linear programming problem having more than two decision variables. Both methods provide a technique for starting from an extreme feasible point within the solution space, and improving upon the solution by proceeding to some other extreme points within the feasible region until the optimal solution to the problem is attained.

The generalized linear programming model

From the sample problems presented so far we can observe a distinct pattern for the general formulation of a linear programming problem. In each problem, decision variables, an objective function, and system (or structural) constraints were defined, which all-together formed the mathematical model (statement) of a real – world – type situation. Rather than dealing with specific cases as we did in our sample problems, it would be better to know the generalized form of linear programming problems. We therefore proceed thus:

Decision Variables

In each specific case, decision variables which denoted a level of activity or quantity produced were defined. For the general model with n decision variables, the variables are defined as follows:

X_1 = quantity or level of activity one

X_2 = quantity or level of activity two

X_3 = quantity or level of activity three

X_j = quantity or level of activity j

X_n = quantity or level of activity n

or simply as

X_j = quantity or level of activity j,

where j = 1,2,- - -, n.

Objective Function

The objective function is an expression that represents the sum total of the contribution of each decision variables in the model toward a particular objective, which may be either maximization or minimization. It is thus expressed as:

Maximize $Z = C_1X_1 + C_2X_2 + - - - + C_jX_j + - - - + C_nX_n$

where,

Z = the total value of the objective function

C_j = the contribution per unit of acivity j, (j = 1,2, - - -, n)

In the case of minimization problem, the objective function is represented as

Minimize $Z = C_1X_1 + C_2X_2 + - - - + C_jX_j + - - - + C_nX_n$

where,

C_j = the cost per unit of activity j (1,2, - - -, n)

and Z is as defined above.

System (or Structural) Constraints

The constraints of a linear programming model represent the limited availability of resources in the problem. If we denote the amount of each of m resources available by b_i (for resource i = 1, 2, …, m),with b_1 being the availability of resource 1 and so on, and define the amount of resource i consumed per unit production of activity j as a_{ij} (j = 1,2, - - -, n), then the constraint equations can be defined as

$$a_{11}x_1 + a_{12}x_2 + + a_{1j}x_j + + a_{1n}x_n \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 + + a_{2j}x_j + + a_{2n}x_n \leq b_2$$

$$a_{i1}x_1 + a_{i2}x_2 + + a_{ij}x_j + + a_{in}x_n \leq b_i$$

$$a_{m1}x_1 + a_{m2}x_2 + + a_{mj}x_j + + a_{mn}x_n \leq b_m$$

$$x_1, x_2, , x_j, , x_n \geq 0$$

The above general relationship shows that all constraints are of $\leq$ type inequalities. However, in real – world – situations, functional constraints can also take the form.

$$a_{i1}x_1 + a_{i2}x_2 + + a_{ij}x_j + + a_{in}x_n \geq b_i$$

and

$$a_{i1}x_1 + a_{i2}x_2 + \quad + a_{ij}x_j + \quad + a_{in}x_n = b_i$$

The above generalized form of linear programming model can be put together as:

Maximize $Z = c_1x_1 + c_2x_2 + \quad + c_jx_j + \quad + c_nx_n$

(or Minimize Z)

Subject to:

$$a_{11}x_1 + a_{12}x_2 + \quad + a_{1j}x_j + \quad + a_{1n}x_n (\leq, =, \geq) b_1$$
$$a_{21}x_1 + a_{22}x_2 + \quad + a_{2j}x_j + \quad + a_{2n}x_n (\leq, =, \geq) b_2$$

$$a_{i1}x_1 + a_{i2}x_2 + \quad + a_{ij}x_j + \quad + a_{in}x_n (\leq, =, \geq)\ b_i$$

$$a_{m1}x_1 + a_{m2}x_2 + \quad + a_{mj}x_j + \quad + a_{mn}x_n (\leq, =, \geq) b_m$$
$$x_1, x_2, \ , x_j, \ , x_n \geq 0$$

By using the concise algebraic form, this can be represented as

Maximize (or Minimize) $Z = \sum_{j=1}^{n} c_jx_j$

Subject to:

$\sum_{j=1}^{n} a_{ij}x_j (\leq, =, \geq) b_i$, for all i

($i = 1,2, \ldots, m$)

$x_j \geq 0$, for all j ($j = 1,2, \ldots, n$)

where Σ denotes summation sign.

Graphical Solution of LP

The graphical method of solving a linear program consists of three steps.

Step 1: Graphing the constraints and identifying the feasible solution space.

Step2: Finding the optimal solution within the feasible solution space provided by the constraints. That is, finding the values of the decision variables that give the best value of the objective function.

NOTE. Step 2 is not really crucial to finding the solution of a linear programming problem graphically. The main purpose of step 2 is to enable us have an idea of what the objective function looks like when graphed in the two – dimensional plane.

For ease of understanding of the graphical method of solution of a linear program, we examine the following problem:

Problem 1

B & G Inc. produces two products that require two components, A and B. A unit of product 1 requires 3 pieces of component A, 2 pieces of component B, and 1 hour of labour. Each unit of product 2 requires 2 pieces of component A, 4 pieces of component B and 1 hour of labour.

Only 18 pieces and 20 pieces of components A and B, respectively, are available for the production of the two products. However, the production manager must utilize at least 2 hours in the production of the two products. Demand forecast also shows that not more than 4 units of product 2 should be produced. The profit contribution of each unit of product 1 is ₦3.00, while a unit of product 2 contributes ₦4.00 to profit. Determine graphically, the product mix of products 1 and 2 that will maximize total profit.
The above problem can be put in a linear programming form as:

Decision variables: Let,

x_1 = number of units of product 1 to be produced.

x_2 = number of units of product 2 to be produced

The mathematical statement of the problem is:

Maximize $Z = 3x_1 + 4x_2$ (Profit, ₦)

Subject to:

$3x_1 + 2x_2 \leq 18$ (Availability of component A)

$2x_1 + 4x_2 \leq 20$ (Availability of component B)

$x_1 + x_2 \geq 2$ (Labour hours)

$x_2 \leq 4$ (Limitation on product 2)

$x_1, x_2 \geq 0$ (Non negativity)

We can now use the steps listed in the graphical method in finding the optimal solution to the linear program as follows:

NOTE: Purpose of graphing the constraints

The main purpose of graphing the constraints is to represent graphically the set of values (x_1, x_2) that satisfy all the constraints. The region satisfying all the constraints is called the "f**easible solution space"**, and it gives all combinations of x_1 and x_2 that are feasible in terms of meeting all the constraints. In order to graph the four constraint inequalities, we relax the inequalities by treating each of them as an equation. By finding two points that fall on each linear equation, a straight line can be drawn through the two points for each constraint.
This is easily achieved by using the points $(0, x_2)$ and $(x_1, 0)$.

Step 1 – Sketching the Constraints

In this step, we represent graphically the set of values (X_1, X_2) that satisfy all the constraints. This set indicates the feasible solution space which comprises of all combinations of X_1 and X_2 that are feasible in terms of meeting all the constraints to the linear programme.

Constraint 1:

The first constraint to the given problem is $3x_1 + 2x_2 = 18$. To graph this constraint, we first transform it to equality type (i.e., an equation) by expressing it as $3x_1 + 2x_2 = 18$. We next determine two points on the line $3x_1 + 2x_2 = 18$, which will enable us to draw the straight line representing this equation through the two determined points by proceeding as follows:

$3x_1 + 2x_2 = 18$

If $x_1 = 0$, then $3(0) + 2x_2 = 18$; this implies that $2x_2 = 18$, and $X_2 = \frac{18}{2} = 9$

Therefore one point that lies on the line $3x_1 + 2x_2 = 18$ is the point with coordinates $(x_1, x_2) = (0, 9)$. Similarly, if $x_2 = 0$, then $3x_1 + 2(0) = 18$

This gives $x_1 = \frac{18}{3} = 6$. So a second point on the line $3x_1 + 2x_2 = 18$ is the point with coordinates $(x_1, x_2) = (6, 0)$.

We next locate each of these two points, that is, (0, 9) and (6, 0) in the appropriate positions in the x_1 and x_2 plane.

N.B. Note that $X_1 = 0$ along the X_2 – axis, and $X_2 = 0$ along the X_1 – axis.

The two points are thus located as shown in the figure below.

A straight line is then used to connect the two points together as shown in the figure. All pairs of values of X_1 and X_2, for any point on the line, satisfy the equation $3X_1 + 2X_2 = 18$.

We observe that this straight line, $3X_1 + 2X_2 = 18$, divides the X_1/X_2 plane into two regions, namely:

(i) the region below the line towards the origin;

(ii) the region above the line towards the open space.

Out of these two regions only one of them satisfies the first inequality constraint $3X_1 + 2X_2 \leq 18$. We need to determine the region that satisfies the constraint and then identify it by shading. The determination of the region is achieved by proceeding as follows:

1. Consider the origin that is the point 0. At the origin, the value of each of X_1 and X_2 is zero.
2. Substituting this values for x_1 and x_2 in the inequality $3X_1 + 2X_2 \leq 18$ yields $3(0) + 2(0) = 0$.

Since 0 (zero) is less than 18, then it is the region below the line $3X_1 + 2X_2 = 18$ towards the origin that satisfies the inequality $3X_1 + 2X_2 \leq 18$. Consequently, for any point considered along the line $3X_1 + 2X_2 = 18$ and below it, the inequality $3X_1 + 2X_2 \leq 18$ is satisfied. This region is thus shaded as shown below.

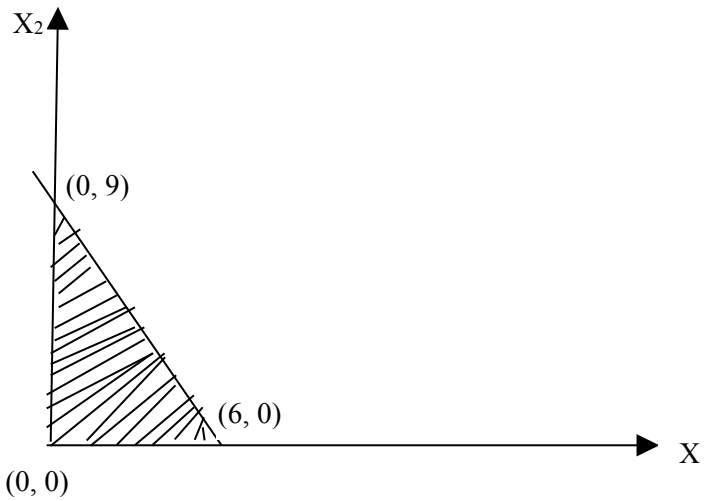


Figure: Feasible region for $3x_1 + 2x_2 \leq 18$

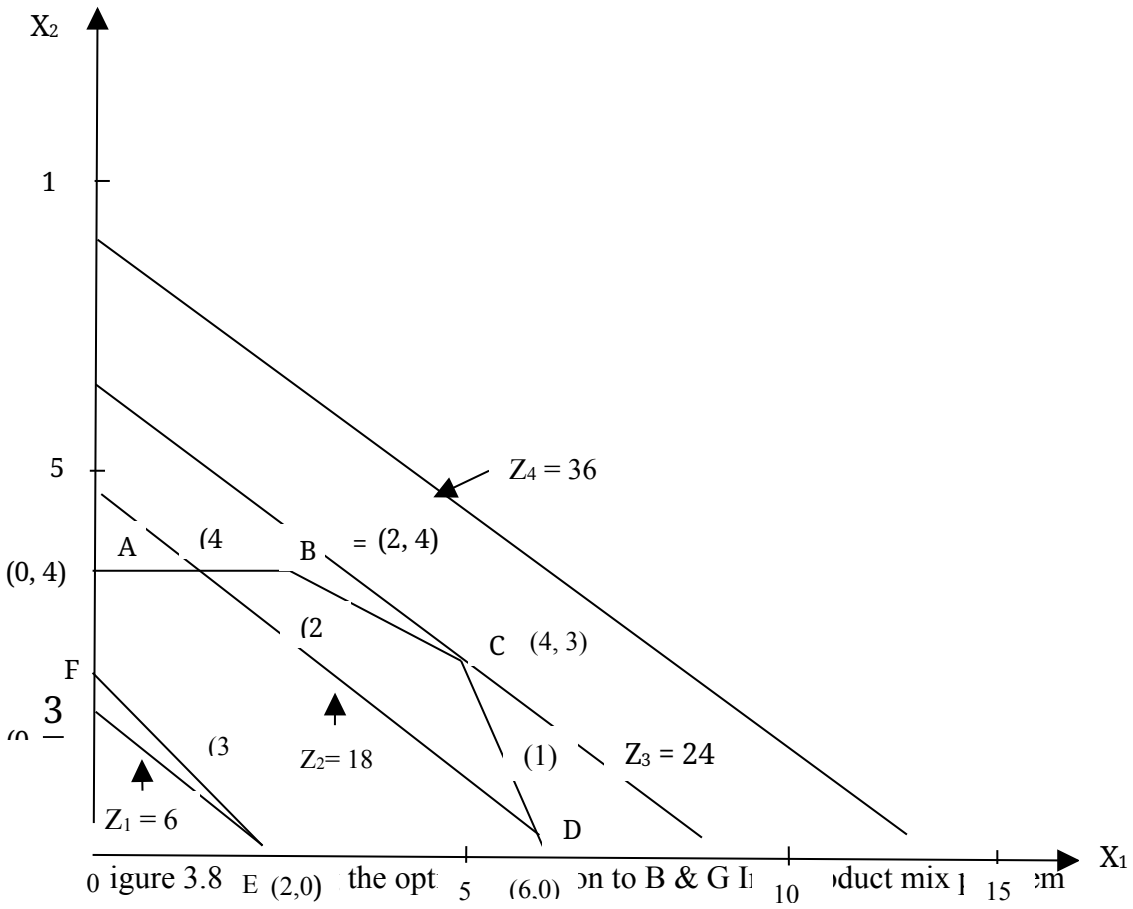
Each of the remaining constraints is treated in a similar way until the graphs of all the constraints have been drawn.

Feasible Solution Space:

The region that satisfies all the constraints simultaneously becomes the feasible solution region that contain the solution to the problem.

Step 3 – Finding the Optimal Solution

Step 3 combines the results of Steps 1 and 2 by combining the feasible solution space in figure 3.6 with those of the objective function in figure 3.7 as illustrated in figure 3.8.



igure 3.8 the opt on to B & G I duct mix m

Since the problem is to make the profit as large as possible, it is clear that Z_1 is not the best value of Z because the objective function can take on larger values as shown by Z_2, Z_3 and Z_4. However, it is equally apparent that Z_4 is not optimal because it contains no points (X_1, X_2) that satisfy the problem constraints (1), (2), (3), and (4) [that is, no point of the line lies within the feasible solution space ABCEF). Z_3, however, has a point C, where $X_1 = 4$ and $X_2 = 3$, in common with the feasible solution space and it is larger than all other Z_j values within the region. The slightest increase in the value of Z beyond the value of Z_3 would cause the objective function line to fall completely out of the feasible solution space. On the other hand, if the value of Z_3 were decreased by a small amount, it would no longer include the optimal point. Thus, the maximum value of Z is located at point C.

Since the optimal solution is found at a '**corner point'** formed by the intersection of two constraints (namely: constraints ((1) and (2)) the (X_1, X_2) coordinates of point C are determined by solving these two constraint equations (that is, availability of components A and B) simultaneously; thus:

$$3X_1 + 2X_2 = 18 \quad (1)$$
$$2X_1 + 4X_2 = 20 \quad (2)$$

Multiplying Equation (1) by 2 yields

$$6X_1 + 4X_2 = 36 \quad (3)$$

Subtract Equation (2) from Equation (3) to have

$$4X_1 = 16$$
$$X_1 = \frac{16}{4} = 4$$

Substitute the value of X_1 in Equation (1) to determine the corresponding value of X_2. This gives

$$3(4) + 2X_2 = 18$$
$$2X_2 = 18 - 12 = 6$$
$$X_2 = \frac{6}{2} = 3$$

Thus the optimal values of X_1 and X_2 are 4units and 3units, respectively. These are shown as the coordinates of the point C in figure 3.8. The interpretation of this result is that 4units of product 1 and 3units of product 2 should be produced to yield a maximum profit of ₦24.

Comments

The following important observations can be made from the above example:

1. The optimal solution (be it maximization or minimization) of a linear programme will always lie on the boundary of the feasible solution space since the feasible solution space forms a convex set of points. That is to say, the boundary of the region is made up of sets of straight lines that converge at corners (often referred to as extreme points). We observe also that a line connecting any two points in the solution space is also in the solution space. Hence as a result of the property of convexity, the boundary formed by the constraint equations must contain the set of points that includes a

unique maximum (or minimum) value of Z. Therefore, the boundary must contain the optimal point.

2. The optimal solution is not only on the boundary of the solution space but more specifically it is at a corner point formed by the intersection of two or more constraints. This is because the corner points are protrusions (or extremes) in the convex set and, thus, the outermost points on the boundary. It, therefore, follows that for any linear programming problem, each extreme point feasible solution is at the intersection of n constraint equations, where n is the number of decision variables. Thus, the solution at any corner point can be found by solving n simultaneous equations. The only exception to this property is in a case of multiple optimal solutions that occurs when the objective function is parallel to a constraint line, that is, when the objective function and one of the constraint equations have the same slope.
3. In a maximization linear programming problem, the corner points that provide the optimal solution is always further away from the origin, whereas in a minimization problem the corner point is always nearer the origin.

The Simplex Method

Introduction

The simplex method of solving linear programming problems was first developed by George B. Dantzig in 1947 and has since been refined by numerous others. The simplex method is based on matrix algebra in that a set of simultaneous constraint equations is solved through the matrix inverse procedure. The mathematical procedure employs an iterative process of repeating a set of mathematical operations, until the optimal solution is attained. In ~~other words, in a minimization problem each successive operation yields a total cost smaller than the cost resulting from the previous operation. In a maximization problem each successive operation yields a higher profit than the previous one~~.

The simplex method is an algorithm for starting at some extreme feasible point and, by a sequence of exchanges, proceeding systematically to other such points until a solution point is found. This is done in a way which steadily reduces (or increases) the value of Z, depending on the type of problem.

Transformation of linear programming model to the standard Simplex form

To employ the simplex method for solving linear programming problems, the inequality constraints in the model must first be transformed to equalities. Since the simplex method is based on the principles of matrix algebra, it entails the solution of sets of simultaneous equations. The constraints of linear programming models, however, are often the inequality types. Because inequalities are difficult to solve simultaneously, simplex method requires that all the inequality constraints be first transformed into equalities by the addition of slack or surplus variables depending on whether it is a maximization or a minimization problem. To demonstrate this transformation, consider the linear programming model.

Maximize $Z = 6X_1 + 5X_2$

Subject to:

	Constraint number
$4X_1 + X_2 \leq 800$	(1)
$2X_1 + 3X_2 \leq 900$	(2)
$X_1 \leq 180$	(3)
$X_2 \leq 320$	(4)

where $X_1 \geq 0$ and $X_2 \geq 0$.

In this problem, the $\leq$ inequality constraints are converted to equation by inserting in each constraint a new variable called slack variable. For this problem, four slack variables are needed; one for each of the four constraints. The addition of these slack variables (denoted by S_i) results in the following converted constraint equations

$$4X_1 + X_2 + S_1 = 800$$
$$2X_1 + 3X_2 + S_2 = 900$$
$$X_1 + S_3 = 180$$
$$X_2 + S_4 = 320$$

The whole problem can now be expressed in its canonical form as:

Maximize $Z = 6X_1 + 5X_2 + 0S_1 + 0S_2 + 0S_3 + 0S_4$

Subject to:

$$4X_1 + 1X_2 + 1S_1 + 0S_2 + 0S_3 + 0S_4 = 800$$
$$2X_1 + 3X_2 + 0S_1 + 0S_2 + 0S_3 + 0S_4 = 900$$
$$1X_1 + 0X_2 + 0S_1 + 0S_2 + 1S_3 + 0S_4 = 180$$
$$0X_1 + 1X_2 + 0S_1 + 0S_2 + 0S_3 + 1S_4 = 320$$
$$X_1, X_2, S_1, S_2, S_3, S_4 \geq 0$$

Notice that the slack variables have zero coefficients everywhere except in the one constraint to which they are applicable. In reality, slack refers to the amount of unused labour if 200units of product 1 and zero units of product 2 are produced. Similar analyses can be performed at other corner points and for all points within the feasible solution space. For any point that is not on the boundaries of one of the constraint within the solution space, slack will be present in all the constraints since all possible resources are not being utilized. At the origin for instance, where both X_1 and $X_2 = 0$ (which is a feasible solution point) $S_1 = 800$, $S_2 = 900$,

$S_3 = 180$ and $S_4 = 320$. This implies that if there is no production, then all of the available resources are unused or slack.

Simplex Method

The process in the simplex solution method consists of three basic steps.

1. The process starts at the origin (i.e., the initial solution is defined as $X_1 = 0$ and $X_2 = 0$
2. After the initial solution is selected, the simplex process searches for a better solution (if it exists), i.e., a solution with a higher profit (or lower cost), Z, value.
   When a better solution is found, the simplex procedure automatically eliminates all other solutions that are not as good from further consideration.

3. This process is repeated until a better solution cannot be found. The simplex process then terminates and the optimal solution is indicated.

The General Simplex Tableau

The conversion of the standard linear programming model into tableau form is one of the major characteristics of the simplex method that contributes to the computational simplicity and ease of understanding. A format of the simplex tableau is presented in the Table below.

Format of Simplex tableau

A	B	C	D				
C_j / $C_b = C_i$	Basis	b_i	C_1	C_2	C_3		C_{n+m}
			X_1	X_2	X_3		
	Z_j						
	$C_j - Z_j$						

The symbol definitions used in the above table are as follow:

Elements in row C_j at the top of the tableau

The C_j is the profit or cost coefficients of the objective function. These are shown for each variable as C_1, C_2, C_3, …, C_{n+m} (where n is the number of decision variables and m the number of constraint equations).

Elements under A

The $C_b = C_i$ [where i indicates the row number and remains constant from one iteration to the other] is set equal to the C_j value, from the objective equation, of the basic variable represented in row *i*. In other words, C_b is the profit or cost coefficients of the objective function for variables in the 'basis" solution.

Elements under B

The "basis" contains the variables currently in the solution set, that is, the variables that have values at the present solution point being analyzed. This is referred to as a basic feasible solution.

Elements under C

The b_i's are a set of values representing the RHS of the constraints at the initial stage. As the simplex method iterates toward a solution, these values are the "solution" values for corresponding variables in the solution basis.

Elements under D

The elements under D can be put in four groups as follows;

(i) X_j, j = 1,2,….,n+m, are the variables. For a maximization problem, all the structural constraints in the original problems are ≤ type, hence we need slack variables X_{n+m} (m = 1,2,….) for each of the constraints. Where some of the constraints are equalities (i.e.=), then we must add artificial variables to permit basic variables in the constraints concerned.

(ii) The box immediately below the X_j row contains a set of coefficient (a_{ij}) which represents the rate of consumption of the ith resource when one unit of the variable named in the column is included in the basis. In other words, the box is the matrix of the coefficient of the variables appearing in the constraint equations.

(iii) The row corresponding to Z_j contains the opportunity cost of introducing the corresponding variable in the column into the basic feasible solution. Thus, Z_j is the opportunity cost of introducing X_j into the basic feasible solution.

(iv) The last row, $C_j - Z_j$, is the difference between the variable's contribution to the objective function and its opportunity cost.

Example 1

We now illustrate the above discussion by using a specific example. Consider the linear programming problem:

Maximize $Z = 40X_1 + 45X_2 + 24X_3$

Subject to:

$$2X_1 + 3X_2 + X_3 \leq 100$$
$$3X_1 + 3X_2 + 2X_3 \leq 120$$
$$X_1, X_2, X_3 \geq 0$$

Canonical Form

The canonical format of this problem is

Max. $Z = 40X_1 + 45X_2 + 24X_3 + 0S_1 + 0S_2$

subject to:

$$2X_1 + 3X_2 + X_3 + S_1 + 0S_2 = 100$$
$$3X_1 + 3X_2 + 2X_3 + 0S_1 + S_2 = 120$$
$$X_1, X_2, X_3, S_1, S_2 \geq 0$$

Table 1: Tableau form of the problem in example 1.

C_j / C_i	Basis	b_i	40	45	24	0	0
			X_1	X_2	X_3	S_1	S_2
$C_1 = 0$	S_1	100	2	3	1	1	0
$C_2 = 0$	S_2	120	3	3	2	0	1
	Z_j	0	0	0	0	0	0
	$C_j - Z_j$		40	45	24	0	0

By simply observing Table 1 and the canonical form of the linear programming problem, the sources of the values in the tableau are evident. To promote better understanding of the

simplex process, however, the origin of the values in the Tableau cell will be explained in greater detail.

The column with the heading basis in the tableau contains the variables that are in the initial solution set. As noted earlier, the first step in the simplex solution procedure is to start with an initial basic solution at the origin and then proceed to better solutions.

Given a linear programming problem with n variables and m constraint equations, there will always be m basic variables and n – m non basic variables. That is to say, basic variables are those variables in the solution basis of the simplex tableau, with associated solution values, while non-basic variables are those variables that are not in the solution basis, and, as such, take the value zero. For the above example, for instance, each iteration of the simplex solution process will contain two basic variables corresponding to the two constraints (that is, m = 2), and there will be three non-basic variables (n – m = 5 – 2 = 3).

From the above analysis, we now know that a basic solution for our problem consists of two basic variables while the other three variables equal zero. The initial basic solution occurs where $S_1 = 100$ and $S_2 = 120$. Thus, the other three decision variables, X_1, X_2 and X_3, are non -basic and hence equal to zero. However, $S_1 = 100$ and $S_2 = 120$ denote respectively that all the labour and material resources are unused. This implies that no productive activity is taking place, hence X_1, X_2 and X_3 must equal zero. When X_1, X_2 and X_3 are non-basic (i.e., equal to zero), the associated solution is at the origin of coordinate. This is a general property of all simplex problems of this nature. The initial solution that begins the simplex process always commences at the origin. As such, the two variables forming the solution base in Table 2 are S_1 and S_2. We also observe that the number of blank cells under the column titled basis in

Table 1 corresponds to the number of constraint equations in the problem and, hence, the number of basic variables in the solution. Thus, the total number of rows in the initial simplex tableau will remain constant for subsequent iterations. Also, because the simplex process in its present form automatically excludes negative values, the non-negativity constraints X_1, X_2, X_3, S_1, $S_2 \geq 0$ are not explicitly shown in the tableau as constraint rows. It should however be noted that this does not mean that conditions where negative values may exist cannot arise.

The b_i column in Table 2 indicates the right-hand-side value of each constraint equation. In the initial tableau, therefore, the b_i values are 100 for S_1 and 120 for S_2.

The C_j values along the top row are the contributions per unit of each variable to the objective function. In the C_j row in Table 2, $C_1 = 40$, $C_2 = 45$, $C_3 = 24$, $C_4 = 0$, $C_5 = 0$. For those variables in the solution basis, the C_j values are also indicated along the extreme left column of the simplex tableau, denoted by C_i. Since S_1 and S_2 are in the initial solution bases, $C_1 = 0$ and $C_2 = 0$ appear in this column.

The X_j and S_i headings refer to the variables in the model. The format of the simplex tableau requires that the decision variables (X_j) be listed first in the order of the magnitude of their subscript (j) followed by the slack variables which are also listed from left to right in order of magnitude of their subscript (i). Thus in 2 the variables read from left to right X_1, X_2, X_3, S_1 and S_2.

The values within the tableau in the columns under each variable (X_1, X_2, X_3, S_1, S_2) are the technological coefficients, a_{ij}, values in each constraint equation. Since the first row

corresponds to the first constraint, the a_{ij} values are $a_{11} = 2$, $a_{12} = 3$, $a_{13} = 1$, $a_{14} = 1$, $a_{15} = 0$ (where a_{14} and a_{15} are respectively the coefficients of the slack variables S_1 and S_2 in the first constraint).

Developing the Initial Simplex Tableau

To proceed with developing the initial simplex tableau, we first reformulate each of the side constraints in terms of their respective slack variables. In our sample problem, for instance, the first and second constraints are restructured in terms of their slack variables as follows:

$S_1 = 100 - 2X_1 - 3X_2 - X_3$

$S_2 = 120 - 3X_1 - 3X_2 - 2X_3$

The objective function is also brought into the first tableau by writing the contribution coefficient of each variable above the variable name. The elements in the C_j row of the tableau have informational value which enables us to determine if we have reached an optimal solution. In the initial tableau, the solution is usually not optimal. For a maximization problem, as long as there are positive values in the $c_j - z_j$ row, under the variables, an optimal solution is yet to be attained; while the values should not be negative in the case of minimization problem. As long as an optimal solution is yet to be attained, the simplex tableau must be revised to improve the solution.

The Simplex Computational Process

The simplex computational process involves basically two steps.

1. Identify the initial basic feasible solution and complete the initial tableau
2. Proceed to better basic feasible solutions until the optimal solution is found.

This second step consists of a number of sub-steps which are enumerated under simplex rules

To complete the initial tableau we need to evaluate the objective function value of the basic feasible solution. This is achieved by computing the Z_j row in the simplex tableau as shown in Table 2.

The values in the Z_j row are determined by multiplying each a_{ij} value in the same column by its corresponding c_b (i.e., c_i) value and then summing these products. For example, in the b_i column the following computation is made:

c_i		b_i	
0	x	100	= 0
0	x	120	= 0
		Z_b	= 0

While in the X_1 column we have

c_i		x_1	
0	x	2	= 0
0	x	3	= 0
		Z_{X1}	= 0

The Z_j value in the b_i column represents the total profit contribution of the basic solution. Since the solution base consists of only slack variables (S_1 and S_2), which contribute nothing to profit, the total value of the objective function is zero.

The last row of the tableau, $C_j - Z_j$ row, is computed by subtracting the Z_j row values from the corresponding C_j values shown in the first row of the tableau (i.e., at the top of the tableau). This row, $C_j - Z_j$, indicates the net increase in profit associated with one additional unit of each variable. That is to say, $C_j - Z_j$ is the per-unit net increase in profit of entering a non-basic variable in the solution basis. For example, in Table 2, the variables X_1, X_2, X_3 are non-basic. If X_1 is to enter the solution basis, then each unit of product 1 will contribute ₦40.00 to the profit, while including X_2 in the solution basis implies that each unit of product 2 will contribute ₦45.00 to the profit. For the basic variables, the $C_j - Z_j$ values are all zero. This means that since these variables are already in the solution, they will neither increase nor decrease the current solution.

Simplex rules for improving the result in the tableau

Now that the initial simplex tableau has been completed, the solution can be improved upon by taking the following steps.

1. Search the $C_j - Z_j$ row. If all terms (entries) are non-positive, optimality has been achieved in the case of a maximization problem. The reverse (i.e., all entries in the $C_j - Z_j$ row are non-negative) is the case for a minimization problem.
2. Select the entering non-basic variable (Pivot Column). The non-basic variable that should enter the solution basis and become basic variable must first be selected.
   The non-basic variable that is to enter the basis is the one that has the largest increase in profit per unit represented by $C_j - Z_j$.
   Using our example in Table 2 as an illustration, the entering variable is X_2, since the $C_j - Z_j$ value of 45 is the greatest net increase of the three non-basic variables. The X_2 column, in Table 4.2 becomes the pivot column. It is called the pivot column because the elements in this column are used to compute the entering and exiting variables for the next tableau.
3. Select the leaving basic variables (pivot row). If there is an entering (non-basic) variable, then one of the variables in the solution basis must leave. This is referred to as the leaving basic variable. The selection of the pivot row determines the variable to exit the solution basis.

   To determine the pivot row only the positive a_{ij} technological coefficients in the pivot column are taken into consideration. Divide the right-hand-side values b_i by their corresponding positive a_{ij} in the pivot column. The one that gives the smallest ratio determines the pivot row. The consideration of only the positive a_{ij} elements appearing in the pivot column is because they indicate the amount by which the slack variables would decrease for each unit increase in the incoming X_j decision variable. The choice of the smallest ratio row is just to make sure that all the right-hand side constraints are satisfied. In effect, we have selected a side constraint (i.e., pivot row) that is more binding than the others, relative to the usage rates of the technological coefficients in the pivot column. The logic is that by considering this constraint early in the solution process, the available resources will be more fully utilized and thus maximize Z. The nonnegative element common to both the pivot column and the pivot row is called the pivot element.

In general, the selection of the pivot row can be expressed symbolically as:

$$\text{Minimum over all } i \quad \left\{\frac{b_i}{a_{ik}}\right\}$$
$$a_{ik} > 0$$

where k denotes the pivot column and i = the row (i = 1, 2, …., m) under consideration.

Also, the entering variable at its entering amount is expressed in terms of symbols as:

$$X_k = \text{Min. over all } i \left\{\frac{b_i}{a_{ik}}\right\}$$
$$a_{ik} > 0$$

Using the example in Table 2 as an illustration,

(a) Select the Pivot Column

The column headed by X_2 is the pivot column since it has the greatest $C_j - Z_j$ value of 45. The decision variable X_2 is to enter the basis.

(b) Select the Pivot Row

The pivot row is given by:

$$\text{Minimum over all } i \left\{\frac{b_i}{a_{i2}}\right\}$$
$$a_{i2} > 0$$

That is,

$$\text{Row with Minimum } \left[\frac{b_i}{a_{12}}, \frac{b_2}{a_{22}}\right]$$
$$a_{i2} > 0$$
$$= \text{min.} \left[\frac{100}{3}, \frac{120}{3}\right] = \text{min.} \left[33\frac{1}{3}, 40\right] \qquad = \text{row 1.}$$

The entering non basic variable and its entering amount is

$$X_2 = \text{min. over all } i \left\{\frac{b_i}{a_{i2}}\right\}$$
$$a_{i2} > 0$$
$$= \text{min.} \left[\frac{b_i}{a_{12}}, \frac{b_2}{a_{22}}\right] = \text{min.} \left[\frac{100}{3}, \frac{120}{3}\right] \qquad = 33\frac{1}{3}$$

4. Select the Pivot Element

The a_{ij} at the point of intersection of the pivot column and the pivot row is called the pivot element. In Table 2 for example,

$$\text{New row value} = 3 - \left(3 \times \frac{2}{3}\right) = 3 - 2 = 1$$

That is column (2) – column (3) x column (4) = column (5)

The computations of all other values are done in the same way. The results are shown in Table 3(a)

Table 3(a) Computation of S_2 row values

(1) Column	(2) Old Row Value		(3) Coefficient in Pivot Column		(4) New Pivot Row Value		(5) New Row value
b_i	120	—	(3	x	$33\frac{1}{3}$)	=	20
X_1	3	—	(3	x	1)	=	1
X_2	3	—	(3	x	$\frac{2}{3}$)	=	0
X_3	2	—	(3	x	$\frac{1}{3}$)	=	1
S_1	0	—	(3	x	1)	=	-1
S_2	1	—	(3	x	$\frac{1}{3}$)	=	1
					0)		

The new row values are shown in Table 3(b).

Table 3 (b) New row values in the Second Simplex Tableau

C_j / C_i	Basis	b_i	40	45	24	0	0
			X_1	X_2	X_3	S_1	S_2
$C_i = 45$	X_2	$33\frac{1}{3}$	$\frac{2}{3}$	3	$\frac{1}{3}$	$\frac{1}{3}$	0
$C_2 = 0$	S_2	20	1	0	1	-1	1
	Z_j						
	$C_j - Z_j$						

In b_i column, the basis value of S_2 (i.e., 20) represents the remaining slack in material when $33\frac{1}{3}$ units of product 2 is produced.

5. The Z_j and $C_j - Z_j$ row values must be determined to complete the second simplex tableau. This is done in a manner similar to how the values were determined in the initial simplex tableau. The values in the Z_j row are computed by summing the products of the values in each column and the corresponding C_i values. Thus,

$Z_{bi} = (33\frac{1}{3} \times 45) + (20 \times 0) = 1{,}500$

$Z_1 = (\frac{2}{3} \times 45) + (1 \times 0) = 30$

$Z_2 = (1 \times 45) + (0 \times 0) = 45$

$Z_3 = (\frac{1}{3} \times 45) + (1 \times 0) = 15$

$Z_4 = (\frac{1}{3} \times 45) + (-1 \times 0) = 15$

$Z_5 = (0 \times 45) + (1 \times 0) = 0$

The $C_j - Z_j$ values are computed by subtracting each Z_j value from the corresponding C_j value (in the same column) at the top row of the tableau. This gives the complete second simplex tableau shown in Table 3(c).

Table 3 (c) Complete Second Simplex Tableau

C_j / C_i	Basis	b_i	40	45	24	0	0
			X_1	X_2	X_3	S_1	S_2
$C_i = 45$	X_2	$33\frac{1}{3}$	$\frac{2}{3}$	1	$\frac{1}{3}$	$\frac{1}{3}$	0
$C_2 = 0$	S_2	20	1	0	1	-1	1
	Z_j	1,500	30	45	15	15	0
	$C_j - Z_j$		10	0	9	-15	0

This completes the first iteration, and the value of the objective function for this basic feasible solution is 1,500.

NOTE 1:

Since we now have different values for the technological coefficients a_{ij}, these values are denoted by $\bar{a}_{ij}$ after the first iteration to distinguish them from the original a_{ij}. Also, the new set of values in the b_i column are now denoted by $\bar{b}_i$.

For any solution after the first iteration, the value of the objective function is given by

$$\text{Objective value} = \sum_{i=1}^{m} C_i \bar{b}_i$$

while the Z_j are computed by setting:

$$Z_j = \sum_{i=1}^{m} C_i \bar{a}_{ij}$$

Third Simplex Tableau

The second iteration that leads to the third simplex tableau begins by repeating the simplex process.

1. Select the entering non-basic variable by searching for the largest positive value in the $C_j - Z_j$ row in the final second simplex tableau. From the table, $C_1 - Z_1$ is the greatest, indicating that X_1 has the greatest per unit net increase in profit. Thus, X_1 is the entering non-basic variable and the column headed by X_1 is the pivot column.

2. Select the leaving basic variable (i.e., pivot row) by comparing the ratios of b_i and the corresponding values of the a_{ij} in the pivot column as follows:

Basis	b_i	X_1	
X_2	$33\frac{1}{3}$	$\frac{2}{3}$	= 50
S_2	20	1	= 20

Minimum over all $i\left[\frac{b_i}{a_{ik}}\right]$ = Minimum over all $i\left[\frac{b_i}{a_{i1}}\right]$

$a_{ik} > 0$ $a_{i1} > 0$

= Minimum [50, 20] = 20

Thus, row 2 becomes the pivot row and S_2 leaves the basis to become non-basic variable. The entering basic variable, X_1, row replaces the slack variable S_2 in the pivot row, that is, row 2. Also, the value at the point of intersection of the pivot column and the pivot row (that is, 1) is now the pivot element. Furthermore, the amount with which X_1 enters the basic feasible solution is minimum (50, 20) = 20.

Performing the row operation on the pivot row by dividing each cell value by the pivot element [1], results in the new pivot row values in the third tableau. The row values for the X_2 row are computed by using:

New row value =	Old row value –	[Corresponding		New pivot]
		[Coefficient in pivot	x	row value]
		[Column along the		along the]
		[Same row		same column]

This yields the values shown in Table 3(d).

Also the Z_j are computed using:

$$Z_j = \sum_{i=1}^{2} C_i a_{ij}$$

The values of the $C_j - Z_j$ are computed in the same manner as in the second tableau .

Table 3 (d) Pivot Column / row and their values.

C_j / C_i	Basis	b_i	40	45	24	0	0
			X_1	X_2	X_3	S_1	S_2
$C_i = 45$	X_2		0				
$C_2 = 40$	X_1	20	1	0	1	-1	1
	Z_j		40				
	$C_j - Z_j$		0				

Table 4: X_2 row values for third simplex tableau

Column	Old Row Value	—	Coefficient in Pivot Column	x	New Pivot Row Value	=	New Row value
b_i	$33\frac{1}{3}$	—	$(\frac{2}{3}$	x	20)	=	20
X_1	$\frac{2}{3}$	—	$(\frac{2}{3}$	x	1)	=	0
X_2	1	—	$(\frac{2}{3}$	x	0)	=	1
X_3	$\frac{1}{3}$	—	$(\frac{2}{3}$	x	1)	=	$-\frac{1}{3}$
S_1	$\frac{1}{3}$	—	$(\frac{2}{3}$	x	-1)	=	1
S_2	0	—	$(\frac{2}{3}$	x	1)	=	$-\frac{2}{3}$

$Z_{b_i} = (20 \times 45) + (20 \times 40) = 1{,}700$

$Z_1 = (45 \times 0) + (40 \times 1) = 40$

$Z_2 = (45 \times 1) + (40 \times 0) = 45$

$Z_3 = (45 \times \left(-\frac{1}{3}\right)) + (40 \times 1) = 25$

$Z_4 = (45 \times 45) + (40 \times (-1)) = 5$

$Z_5 = (45 \times \left(-\frac{2}{3}\right)) + (40 \times 1) = 10$

Table 5 Completed third Simplex tableau

C_j / C_i	Basis	b_i	40	45	24	0	0
			X_1	X_2	X_3	S_1	S_2
$C_i = 45$	X_2	20	0	1	$-\frac{1}{3}$	1	$-\frac{2}{3}$
$C_2 = 40$	X_1	20	1	0	1	-1	1
	Z_j	1,700	40	45	25	5	10

	$C_j - Z_j$		0	0	-1	-5	-10

Observing the $C_j - Z_j$ row, we see that no positive value exists. X_1 and X_2 being basic variables have zero $C_j - Z_j$ values, while there are negative $C_j - Z_j$ values for the three non-basic variables (i.e., X_3, S_1 and S_2). The negative $C_j - Z_j$ values imply that if any of them is selected as the entering variable, the objective function value would decrease by the corresponding amount. That is to say, if S_2 for instance, is injected into the basis, the objective function value would decrease by 10 for every unit of S_2 that enters the basis.

The fact that there are no positive $C_j - Z_j$ values indicates that there are no non-basic variables that could enter the solution to further increase profit. Therefore, the simplex process terminates since an optimal solution has been reached. The value of the objective function at the optimal solution is 1,700, while 20 units each of product 1 and product 2 must be produced to make this value. Therefore,

$X_1^* = 20$, $X_2^* = 20$, $X_3^* = 0$, $s_1^* = 0$, $s_2^* = 0$ and value of the objective $Z(X^*) = 1{,}700$.

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NOTE: That the a_{ij} s for the variables in the basic feasible solution constitutes a unit vector with a 1 in the row in which the variable serves as a basis and zero elsewhere.

Example 2: Product Mix Problem

A manufacturing firm produces two products, 1 and 2. The requirements for labour and materials for production of each product as well as the resource availabilities are as follows:

	Resource Requirement		
Resource	Product 1	Product 2	Available Resources
Material	31b/unit	51b/unit	Not more than 20 1b
Labour	2hr/unit	6hr/unit	At least 30hr

The unit profit for product 1 is ₦4 and for product 2, ₦5. Management problem is to determine the optimal product mix that will maximize profit subject to availability of the limited resource.

The linear programming model for this problem is formulated as

Maximize $Z = ₦4X_1 + ₦5X_2$ (Profit)

Subject to

$3X_1 + 5X_2 \leq 20$ (Material)

$2X_1 + 6X_2 \geq 30$ (Labour)

$X_1, X_2 \geq 0$

where

X_1 = number of units of product 1 to produce

X_2 = number of units of product 2 to produce

Z = total profit (₦)

The problem is transformed into its standard form by converting the $\leq$ and $\geq$ inequalities

to equations by inserting in each constraint a new variable called a slack variable. The addition of these slack variables denoted by s_i, results in the following converted constraint equations:

$3X_1 + 5X_2 + S_1 = 20$ (Material)

$2X_1 + 6X_2 + 0S_1 - S_2 = 30$ (Labour)

,where

S_1 = slack variable for material constraint

S_2 = slack variable for labour constraint

While this converted problem is in standard form, it is not in canonical format since s_2 cannot be an initial basic variable for the second constraint equation because it has a negative coefficient (i.e., -1). The problem is converted into canonical format by introducing an artificial variable, A, in the labour constraint to give:

Maximize $Z = 4X_1 + 5X_2 + 0S_1 + 0S_2$

Subject to:

$3X_1 + 5X_2 + S_1 + 0S_2 + 0A_1 = 20$ (Material)

$2X_1 + 6X_2 + 0S_1 - S_2 + A_1 = 30$ (Labour)

$X_1, X_2, S_1, S_2, A_1 \geq 0$

It should be noted that when an artificial variable is introduced in a constraint equation the coefficient of the artificial variable in the objective function cannot be zero. Hence the artificial variable must not exist or appear in the optimum solution. To overcome the problem of the artificial variable appearing in the optimal tableau, we assign a constant M of very large magnitude as its coefficient in the objective function.

Thus the problem becomes:

Maximize $Z = 4X_1 + 5X_2 + 0S_1 + 0S_2 - MA_1$

subject to:

$3X_1 + 5X_2 + S_1 + 0S_2 + 0A_1 = 20$

$2X_1 + 6X_2 + 0S_1 - S_2 + A_1 = 30$

$X_1, X_2, S_1, S_2, A_1 \geq 0$

The simplex process can then be applied to obtain the optimal solution of this problem as we did in example 1.

Summary of the Simplex Process

For a maximization problem, the simplex process can be summarized in step wise format as follows:

1. Transform the problem into canonical simplex form. This requires the conversion of the inequality constraints into equations by adding slack and/or artificial variables as required.
2. Set up the initial simplex tableau. The initial basic feasible solution is at the origin. Thus, the initial solution base consists only of slack variables. The values in the Z_j and $C_j - Z_j$ row must be computed.
3. Determine the pivot column (i.e., the entering non-basic variable). This is found by observing the $C_j - Z_j$ row and selecting the largest positive value.

4. Determine the pivot row (i.e., the leaving basic variable). This is done by taking the ratio of the value in the b_i column to the corresponding pivot column value (i.e., the a_{ij}). Select as the leaving variable the one with the minimum nonnegative value (being the most constraining).
5. Compute the new tableau row values. The new row values corresponding to the old pivot row are computed using the formula:

New pivot row value = $\frac{\text{Old pivot row value}}{\text{Pivot element,}}$

While all other row values are computed using the formula:

New row value = Old row value – [Corresponding [Coefficient in pivot [Column along the [Same row x New pivot] row value] along the] same column]

The Z_j are computed using the relation:

$$Z_j = \sum_i C_i a_{ij}$$

The value in the $C_j - Z_j$ row is determined by simply subtracting Z_j from C_j.

6. Confirm if the new solution is optimal. The necessary condition for the new basic solution to be optimal is that all values in the $C_j - Z_j$ row be non-positive (i.e., the values are either zero or negative). If a positive value exists, go to step 3 and repeat the steps in the simplex process.

Self-assessment questions

1. Determine the optimum solution of the LP:

Maximize : $Z = 2x_1 - 4x_2 + 5x_3 - 6x_4$

Subject to

$$x_1 + 4x_2 - 2x_3 + 8x_4 \leq 2$$

$$-x_1 + 2x_2 + 3x_3 + 4x_4 \leq 1$$

$$x_1, x_2, x_3, x_4 \geq 0$$

2. Use the simplex method to solve the following problems:.

(i) Maximize: $Z = 4X_1 + 3X_2 + 6X_3$

subject to $3X_1 + X_2 + 3X_3 \leq 30$

$2X_1 + 2X_2 + 3X_3 \leq 40$

and $X_1 \geq 0,\ X_2 \geq 0,\ X_3 \geq 0.$

(ii) Maximize: $Z = X_1 + 2X_2 + 4X_3$

subject to $3X_1 + X_2 + 5X_3 \leq 10$

$X_1 + 4X_2 + X_3 \leq 8$

$2X_1 + 2X_3 \leq 37$

and $X_1 \geq 0,\ X_2 \geq 0,\ X_3 \geq 0.$

(iii) Maximize: $Z = X_1 + 2X_2 + 2X_3$
subject to $5X_1 + 2X_2 + 3X_3 \leq 15$
$X_1 + 4X_2 + 2X_3 \leq 12$
$2X_1 + X_3 \leq 8$
and $X_1 \geq 0,\ X_2 \geq 0,\ X_3 \geq 0.$

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